Mahesh Chitakoti

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Summary

Backend & AI Engineer with 2.5+ years of experience building scalable microservices, production RAG pipelines, and agentic workflows using **Python (FastAPI), TypeScript (Node.js), PostgreSQL, and Redis**. Experienced in stateful LLM orchestration with **LangGraph, Qdrant, and Pydantic**, with a proven track record of reducing inference costs by 35%, preventing API rate throttling, and tuning p95 query latency.

Technical Skills

Languages & Backend: Python, TypeScript, SQL, FastAPI, Node.js, Express.js, AsyncIO, REST APIs, WebSockets

AI & Retrieval Systems: RAG, Agentic Workflows (LangGraph), Qdrant, pgvector, Hybrid Search (Dense + BM25), Semantic Caching, Pydantic Structured Outputs, Prompt Evaluation

Databases & Cloud Infrastructure: PostgreSQL, Redis, Docker, AWS (EC2, S3), GitHub Actions CI/CD, Gemini APIs, OpenAI APIs

Experience

AI Developer – SahioNext Technology Pvt. Ltd. — Oct 2025 – Apr 2026

Remote

- Built the core LLM evaluation engine using **FastAPI and Gemini APIs**, integrating multi-step **LangGraph agentic workflows with Pydantic validation** and fallback parsers to eliminate broken JSON responses.
- Reduced recurring inference expenses by **35%** and eliminated 429 throttling errors by designing a **Redis semantic caching layer** coupled with token-bucket rate limiters.
- Engineered a hybrid retrieval pipeline (**dense embeddings + BM25**) over 10,000+ lesson transcript chunks in **Qdrant**, achieving **¡300ms p95 latency** and improving keyword precision for vocabulary rubrics.
- Developed bidirectional **WebSocket streaming endpoints** for real-time speech evaluation, implementing client reconnection recovery and state buffering to maintain reliable streams over unstable networks.

Full Stack Developer (Contract) – PearlThoughts — Feb 2025 – Sep 2025

Remote

- Developed multi-tenant workflow automation microservices in **TypeScript and Node.js**, processing **1,000+ daily business actions** across SaaS APIs and eliminating manual data entry bottlenecks.
- Engineered distributed background workers with **idempotent webhook handlers and automatic retry queues**, preventing duplicate payment records and state corruption across concurrent client requests.
- Constructed isolated **Docker sandbox environments** for automated code grading on enterprise Coursera launches, enforcing strict execution timeouts and memory quotas to prevent execution crashes.

Full Stack Developer – The DevOps Team — Oct 2023 – Jan 2025

Bangalore, India

- Profiled bottleneck queries and optimized **PostgreSQL indexing** alongside **Redis caching**, reducing median API response times from **800ms to 320ms (60% decrease)**.
- Redesigned candidate search queries using multi-column compound indexes, cutting database query latency and improving recruiter role-matching discovery rates by **22%**.
- Built responsive onboarding and profile workflows in **Next.js**, running A/B experiments on multi-step forms to lift user registration conversion by **18%**.

Projects

WorkBridge – AI Collaboration Platform — *May 2026 – Present*

Python, FastAPI, Node.js, PostgreSQL, pgvector, Redis, Next.js | Live Project

- Architected a decoupled microservices platform pairing a **Node.js API gateway** for session auth and RBAC with an asynchronous **FastAPI service** handling vector calculations and similarity search.
- Built candidate-to-project recommendation matching with **pgvector HNSW indexing**, executing sub-100ms cosine similarity retrieval against structured skill embeddings.
- Conducted stress testing via **k6**, tuning PostgreSQL connection pooling and Redis session caching to sustain **¡150ms p95 latency under concurrent load**.

Certifications

Oracle Agentic AI Certified Foundations Associate – Oracle — Aug 2026
Oracle AI Vector Search Certified Professional – Oracle — Sep 2025
Machine Learning I – Columbia University — Aug 2025
Oracle Cloud Infrastructure Generative AI Professional – Oracle — Aug 2024

Education

Adichunchanagiri Institute of Technology (VTU) — Chikkamagaluru, KA, India
Bachelor of Engineering — *Aug 2023*